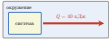


Isotermal expansion, $T = 300\text{ K}$



$$\Delta S_{\text{exp}} = 130\text{ J/K}$$

$$\Delta S_{\text{exp}} = Q/T = 40\,000\text{ J/K} / 300\text{ K}$$

The gas molecules disperse throughout
the container

Isotermal expansion, $T = 600\text{ K}$



$$\Delta S_{\text{exp}} = 60\text{ J/K}$$

$$\Delta S_{\text{exp}} = Q/T = 40\,000\text{ J/K} / 600\text{ K}$$

The gas molecules are not as dispersed
throughout